EVERY ULTRARING IS EQUIVALENT TO THE ULTRARING PRESENTED BY ITS CANONICAL ANNULAR THEORY

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Allowed standard prerequisites (editorial).

- Symmetric monoidal categories, coproducts, Frobenius structures and natural transformations. Author §2 definitions and complete source bibliography retained; all ultraring-specific constructions and source-local proof dependencies included.

2 Ultrarings

2.1 Main Definitions

In what follows, we often abbreviate composition of arrows $f \circ g$ by fg. If $(\mathcal{C}, \otimes, I, \gamma)$ is a symmetric monoidal category, where γ is its symmetry, we denote by $\beta : I \otimes I \to I$ the canonical isomorphism. By "symmetric monoidal functor" we will always mean strong symmetric monoidal functor (i.e., tensor and unit are preserved up to iso).

A Frobenius sesquimonoid in a symmetric monoidal category is an object A equipped with arrows $\delta : A \to A \otimes A$, $\varepsilon : A \to I$, $\delta^* : A \otimes A \to A$ such that (A, δ, ε) is a commutative comonoid, δ^* is commutative ($\delta^* \gamma_{A,A} = \delta^*$), associative ($(\delta^*(\mathrm{id} \otimes \delta^*) = \delta^*(\delta^* \otimes \mathrm{id})$, left inverse to δ ($\delta^*\delta = \mathrm{id}_A$) and verifies the Frobenius law $(\mathrm{id}_A \otimes \delta^*)(\delta \otimes \mathrm{id}_A) = \delta\delta^* = (\delta^* \otimes \mathrm{id}_A)(\mathrm{id}_A \otimes \delta)$.

Initial objects will be denoted by 0. Recall that a (binary) coproduct diagram over an object A is a pair of arrows $\iota_1 : A_1 \to A$, $\iota_2 : A_2 \to A$, called injections, such that, for every $f_1 : A_1 \to B$, $f_2 : A_2 \to B$ there is a unique arrow $[f_1, f_2] : A \to B$ such that $[f_1, f_2]\iota_i = f_i$. In that case, we write $A = A_1 \oplus A_2$.

A distributive monoidal category is a symmetric monoidal category with finite coproducts such that, for all objects A, B, C, $A \otimes 0$ is initial and the arrow $\chi_{A,B,C} := [\mathrm{id}_A \otimes \iota_1, \mathrm{id}_A \otimes \iota_2] : (A \otimes B) \oplus (A \otimes C) \to A \otimes (B \oplus C)$ is an isomorphism.

Recall that an initial object which is also terminal is called a zero object and that any category with a zero object has, for any pair of objects A, B, a zero morphism $0_{A,B} : A \to B$ obtained by composing the terminal and initial arrows $A \to 0 \to B$. Also, for any pair of objects A_1, A_2, the category has coinjections[222]We refer to these as coinjections rather than projections to avoid the reader assuming that the coproducts are also products. defined by:

$\iota_1^* := [\mathrm{id}_{A_1}, 0_{A_2,A_1}] : A_1 \oplus A_2 \to A_1$, $\iota_2^* := [0_{A_1,A_2}, \mathrm{id}_{A_2}] : A_1 \oplus A_2 \to A_2$. It immediately follows that these verify

$$\iota_i^* \iota_j = \begin{cases} \mathrm{id}_{A_i} & \text{if } i = j \\ 0_{A_j,A_i} & \text{if } i \neq j. \end{cases}$$

(1)

Definition 1.

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A δ^*-category is a distributive monoidal category $(\mathcal{C}, \otimes, I, \gamma)$ with a zero object such that each object A is equipped with a Frobenius sesquimonoid structure $\delta_A, \varepsilon_A, \delta_A^*$ satisfying, for every objects A, B, A_1, A_2,

$\delta_{A\otimes B} = (\mathrm{id}_A \otimes \gamma_{A,B} \otimes \mathrm{id}_B)(\delta_A \otimes \delta_B)$ $\varepsilon_{A\otimes B} = \beta(\varepsilon_A \otimes \varepsilon_B)$ $\delta_I = \beta^{-1}$, $\delta^*_{A\otimes B} = (\delta^*_A \otimes \delta^*_B)(\mathrm{id}_A \otimes \gamma_{A,B} \otimes \mathrm{id}_B)$ $\varepsilon_I = \mathrm{id}_I$ $\delta_I^* = \beta$, $\delta^*_{A_1\oplus A_2}(\iota_i \otimes \iota_j) = \begin{cases} \iota_i \delta^*_{A_i} & \text{if } i = j \\ 0_{A_i\otimes A_j, A_1\oplus A_2} & \text{if } i \neq j \end{cases}$ $\delta_{A_1\oplus A_2}\iota_i = (\iota_i \otimes \iota_i)\delta_{A_i}$ $\varepsilon_{A_1\oplus A_2}\iota_i = \varepsilon_{A_i}$, and such that, for all $f, g : A \rightrightarrows B$, $\delta_B^*(f \otimes \mathrm{id}) = \delta_B^*(g \otimes \mathrm{id})$ implies $f = g$.

A morphism $F : \mathcal{C} \to \mathcal{D}$ of δ^*-categories is a finite-coproduct-preserving symmetric monoidal functor with structural isomorphisms $\varphi_{A,B} : FA \otimes FB \to F(A \otimes B)$ and $\psi : I \to FI$ such that, for each object A of $\mathcal{C}$, $F\delta_A = \varphi_{A,A}\delta_{FA}$, $F\delta_A^* = \delta^*_{FA}\varphi^{-1}_{A,A}$ and $F\varepsilon_A = \psi\varepsilon_{FA}$.

In the sequel, subscripts such as those in $\mathrm{id}_A, \delta_A, \varepsilon_A \ldots$ will be dropped when unambiguously retrievable from the context.

Lemma 2.

In a δ^*-category, coinjections are jointly monic.

Proof.

See §A.1. ◀

By the above lemma, for every arrows $f_1 : B \to A_1$, $f_2 : B \to A_2$ of a δ^*-category, there is at most one arrow $\langle f_1, f_2\rangle : B \to A_1 \oplus A_2$, which we call pairing, such that $\iota_i^*\langle f_1, f_2\rangle = f_i$. Some pairings automatically exist. For instance, by (1), $\iota_1 : A_1 \to A_1 \oplus A_2$ is the pairing $\langle \mathrm{id}_{A_1}, 0_{A_1,A_2}\rangle$ and dually for ι_2. The pairing $\langle 0_{B,A_1}, 0_{B,A_2}\rangle$ is simply $0_{B,A_1\oplus A_2}$. More subtly, the inverse χ^{-1}_{A,B_1,B_2} of the distributivity isomorphism is the pairing $\langle \mathrm{id}_A \otimes \iota_1^*, \mathrm{id}_A \otimes \iota_2^*\rangle$, which can be deduced using the fact that injections into a coproduct are jointly epic.

Definition 3.

An ultraring is a small δ^*-category such that:

1.

disjoint pairings: for every $f : C \to A$ and $g : C \to B$, if $(f \otimes g)\delta = 0$ then their pairing $\langle f, g\rangle$ exists;

2.

complements: for every $p : A \to I$ such that $(p \otimes p)\delta = p$, there exist a unique $\overline{p} : A \to I$ such that $(p \otimes \overline{p})\delta = 0$ and $\nabla\langle p, \overline{p}\rangle = \varepsilon$, where $\nabla : I \oplus I \to I$ is the codiagonal.

A morphism of ultrarings is just a morphism of δ^*-categories. We denote by **URing** the strict 2-category whose objects are ultrarings, whose arrows are morphisms between them and whose 2-cells are monoidal natural transformations.

2.2 Examples of Ultrarings

Let R be a commutative ring. We denote by $\mathcal{U}R$ the category of free R-modules of finite rank, and linear maps between them. This may be equivalently described as the category whose objects are natural numbers and whose arrows $n \to m$ are $m \times n$ matrices (m rows, n columns) with coefficients in R, and where composition is matrix multiplication.

This category is distributive monoidal: the (strict) monoidal structure is given by multiplication of natural numbers for objects and by the standard tensor product of matrices (also known as Kronecker product) for morphisms. Coproduct is addition and 0 is easily seen to be a zero object. The Frobenius sesquimonoid structure is as follows: δ_n is the $n^2 \times n$ matrix whose i-th column has the element 1 at position $n(i-1)+i$ and is zero everywhere else; ε_n is the row matrix $(1 \ldots 1)$; and δ_n^* is the transpose of δ_n.

Under matrix addition, $\mathcal{U}R$ is actually an additive category, which means it has biproducts, so point (1) of Definition 3 is immediate. For what concerns point (2), any $p : n \to 1$ is a row matrix $(p_1 \dots p_n)$, and the condition $(p \otimes p)\delta = p$ amounts to $p_i^2 = p_i$ for all i. It is then a matter of elementary calculations to show that $\overline{p} := (1 - p_1 \dots 1 - p_n)$ is the unique morphism $n \to 1$ verifying $(p \otimes \overline{p})\delta = 0$ and $\nabla\langle p, \overline{p}\rangle = \varepsilon$. So $\mathcal{U}R$ is an ultraring.

A ring homomorphism $h : R \to S$ induces a morphism of ultrarings $\mathcal{U}h : \mathcal{U}R \to \mathcal{U}S$ by applying h pointwise to matrices. It is well known (see for instance [7]) that every symmetric monoidal, finite-coproduct-preserving functor $\mathcal{U}R \to \mathcal{U}S$ is monoidally isomorphic to a functor of the form $\mathcal{U}h$ for some $h : R \to S$. Therefore, if we consider the category of commutative rings **CRing** to be a 2-category in which every hom-category is discrete, this defines a 2-functor $\mathcal{U} : \mathbf{CRing} \to \mathbf{URing}$, and we have:

Proposition 4.
$\mathcal{U} : \mathbf{CRing} \hookrightarrow \mathbf{URing}$ is fully faithful.

Let $\mathbf{Fin}_*$ be the category of finite sets and partial functions. Observe that the cartesian product induces a symmetric monoidal structure on $\mathbf{Fin}_*$ (which is not a categorical product), whose unit is the singleton set $\{*\}$, and that $\mathbf{Fin}_*$ is obviously distributive (coproducts are disjoint unions) and has a zero object (the empty set). A Frobenius sesquimonoid structure may be defined by letting $\delta_A : A \to A \times A$ be the diagonal, $\varepsilon_A : A \to \{*\}$ be the unique total function and $\delta_A^* : A \times A \to A$ be the function which is defined on (a, a') exactly when $a = a'$.

We invite the reader to check that, if $f : C \to A$ and $g : C \to B$ are partial functions, the condition $(f \otimes g)\delta = 0$ amounts to the domains of f and g being disjoint. In that case, the pairing $\langle f, g\rangle : C \to A \oplus B$ is the evident partial function whose domain is the union of the domains of f and g. Furthermore, observe that any partial function $p : A \to \{*\}$ satisfies $(p \otimes p)\delta = p$, and corresponds to a subset $P \subseteq A$. The partial function $\overline{p} : A \to \{*\}$ corresponding to the complement of P is easily seen to be the unique one verifying the desired properties of point (2) of Definition 3.

The above shows that $\mathbf{Fin}_*$ has the structure of an ultraring, except that it is not small. This is easily fixed: let $\mathbb{F}_1$ be the category whose objects are natural numbers and whose arrows $n \to m$ are partial functions $\{1, \dots, n\} \to \{1, \dots, m\}$, or equivalently, $m \times n$ matrices with coefficients in $\{0, 1\}$ such that 1 appears at most once in each column. As a skeleton of $\mathbf{Fin}_*$, $\mathbb{F}_1$ is an ultraring. In fact, it is the initial ultraring: since $\mathbb{F}_1$ is generated under finite coproducts by the monoidal unit 1 and its identity arrow, any finite-coproduct-preserving functor $F : \mathbb{F}_1 \to \mathcal{R}$ with $\mathcal{R}$ an ultraring is determined by the image of 1; since morphisms of ultrarings are monoidal, $F1 \cong I$ and $\mathbf{URing}(\mathbb{F}_1, \mathcal{R})$ is equivalent to the terminal category.

Observe that $\mathbf{Fin}_*$ is the Kleisli category of the partiality monad $- + \{*\}$ over the category **Fin** of finite sets and total functions. This is a special case of a more general situation. A category is lextensive if it has finite limits and disjoint, pullback-stable finite coproducts. A lextensive category is called Boolean if every subobject has a complement; see [8]. A primordial example is **Fin**. In fact, for what concerns finite limits and finite coproducts, the objects of Boolean lextensive categories may be thought of as sets and their arrows as total functions. In particular, if $\mathcal{B}$ is a Boolean lextensive category whose terminal object is 1, the endofunctor $- + 1$ is a monad and the Kleisli category $\mathcal{B}_*$ may again be seen as a category of "sets and partial functions". The Kleisli categories of the partiality monad on small Boolean lextensive categories may be characterized as certain ultrarings, which we introduce below.

Definition 5.

An ultraring is Boolean if every arrow f satisfies $\delta f = (f \otimes f)\delta$ and if for every $p : A \to I$ there exists a unique coproduct diagram $(\iota_p, \overline{\iota}_p)$ over A such that $p = \varepsilon \iota_p^*$, $\overline{\iota}_p = \iota_{\overline{p}}$ and, for every arrow f, $\iota_{\varepsilon f} = (\mathrm{id} \otimes \varepsilon)\iota_{\varepsilon\delta^*(f\otimes \mathrm{id})}$.

In the following, we denote by **BoolURing** the full sub-2-category of **URing** on Boolean ultrarings, and by **BoolLext** the 2-category of small Boolean lextensive categories, finitely continuous functors which preserve finite coproducts, and natural transformations.

Proposition 6.
The categories **BoolURing** and **BoolLext** are equivalent.

Proof.
We omit this proof due to space constraints. ◀

3 Annular Theories and Models

3.1 Presentations of Ultrarings

Having introduced the motivating examples of ultrarings, we now wish to provide presentations of these objects, which will subsequently enable us to perform essential constructions on them, as well as providing the basis for a syntax having a natural semantics in ultrarings.

Definition 7.
A signature $\mathfrak{S}$ is a pair $(\mathrm{Sort}(\mathfrak{S}), \mathrm{Gen}(\mathfrak{S}))$ such that:

■

$\mathrm{Sort}(\mathfrak{S})$ is a set of sorts. A tensor of arity n is a sequence of n sorts $A_1 \otimes \cdots \otimes A_n$. The unique tensor of arity 0 is denoted by I.

■

$\mathrm{Gen}(\mathfrak{S})$ is a set of generators, each with a type $g : T \to \bigoplus_{j=1}^{k} U_j$ where T, U_j are tensors.

We fix a countably infinite set of variables, ranged over by x, y, z... and we use $\vec{x}, \vec{y}$,...to denote finite sequences of variables. Given a signature $\mathfrak{S}$, the monomials over $\mathfrak{S}$ are defined as follows:

$$p, q ::= (x =_A y) \mid (g^j(\vec{x}) = \vec{y}) \mid 1 \mid \overline{p} \mid pq \mid \textstyle\int_x p,$$

where A ranges over $\mathrm{Sort}(\mathfrak{S})$, $g : T \to \bigoplus_{j=1}^{k} U_j$ ranges over $\mathrm{Gen}(\mathfrak{S})$, $1 \leq j \leq k$ and the length of $\vec{x}$ and $\vec{y}$ matches that of T and U_j, respectively. We write $g^j(\vec{x})$, $(g^j = \vec{y})$ and g^j rather than $(g^j(\vec{x}) =)$, $(g^j() = \vec{y})$ and $(g^j() =)$, respectively. We abbreviate $(x_1 =_{A_1} y_1) \cdots (x_n =_{A_n} y_n)$ to $(\vec{x} =_{\vec{A}} \vec{y})$. The notation $\int_x a$ is a binder: x is bound in a and is subject to the usual renaming conventions. If $\vec{x} = x_1, \ldots, x_n$, we write $\int_{\vec{x}} a$ for $\int_{x_1} \cdots \int_{x_n} a$. When $\vec{x}$ is empty, $\int_{\vec{x}} a$ is just a. We call a monomial positive if it contains no instances of the negation operator $\overline{p}$.

Monomials generalize logical formulas. Intuitively, in the Boolean case, the monomials 1, $\overline{p}$, pq and $\int_x p$ could be written $\top$, $\neg p$, $p \wedge q$, $\exists x.p$, and the monomial $(g^j(\vec{x}) = \vec{y})$ corresponds to a relation symbol $g^j(\vec{x}, \vec{y})$ satisfying that $g^j(\vec{x}, \vec{y}) \wedge g^j(\vec{x}, \vec{z})$ implies $\vec{y} = \vec{z}$, i.e., a functional relation. The equality monomial $(x =_A y)$ is already a standard Boolean formula.

A pre-polynomial is a formal finite sum of monomials, with 0 representing the empty sum. We can extend the operations of multiplication and integration in the definition of monomials (the last two operations) to pre-polynomials by assuming distributivity and linearity, respectively.

A context is a finite list of type assignments of the form $x : A$ where x is a variable and A a sort, such that no variable appears twice. If $\vec{x} = x_1, \ldots, x_n$ is repetition-free and $T = \bigotimes_{i=1}^{n} A_i$, we also write the context $x_1 : A_1, \ldots, x_n : A_n$ as

$\vec{x} : T$. In particular, in $\vec{x} : I$ the sequence $\vec{x}$ is empty. If Γ and Δ are contexts, Γ, Δ denotes their concatenation (the variables appearing in each are assumed disjoint).

$$\frac{i \in \{1,2\}}{x_i : A \vdash (x_1 =_A x_2) :: x_{3-i} : A} \qquad \frac{g : T \to \bigoplus_{j=1}^{k} U_j \quad 1 \le j \le k}{\vec{x} : T \vdash (g^j(\vec{x}) = \vec{y}) :: \vec{y} : U_j} \qquad \frac{}{\vdash 1 ::} \qquad \frac{\Gamma \vdash p :: \Delta}{\Gamma, x : A \vdash p :: \Delta}$$

$$\frac{\Gamma, x : A, y : A \vdash p :: \Delta}{\Gamma, x : A \vdash p[x/y] :: \Delta} \qquad \frac{\Gamma \vdash p :: \Delta, \vec{x} : T \quad \vec{x} : T, \Gamma' \vdash q :: \Delta'}{\Gamma, \Gamma' \vdash \int_{\vec{x}} pq :: \Delta, \Delta'} \qquad \frac{\Gamma \vdash p :: \Delta, x : A}{\Gamma, x : A \vdash p :: \Delta} \qquad \frac{\Gamma \vdash p ::}{\Gamma \vdash \overline{p} ::}$$

Figure 1: Monomials in context. We omit a rule allowing to permute type assignments in contexts.

A monomial-in-context is an expression of the form $\Gamma \vdash p :: \Delta$, where p is a monomial and Γ and Δ are contexts, which is derivable by means of the inductive rules of Fig. 1. If $\Gamma \vdash p :: \Delta$ is derivable, we say that p is well-typed, that Γ is a domain for p and that the context Γ, Δ matches p. Note that, thanks to the penultimate rule of Fig. 1, any context matching p is also a domain for it. Pre-polynomials-in-context are defined similarly, adding a rule to derive $\Gamma \vdash \sum_{i \in I} p_i :: \Delta$ whenever $\Gamma \vdash p_i :: \Delta$ is derivable for all $i \in I$ (in particular, $\Gamma \vdash 0 :: \Delta$ is derivable for any Γ, Δ). From now on, we will only consider well-typed monomials and pre-polynomials.

$\int_x \int_y p \approx_\Gamma \int_y \int_x p$ $\int_y (x = y) \approx_{\Gamma, x:A} 1$ $(x =_A y)p \approx_{\Gamma, x:A, y:A} (x =_A y)p[x/y]$ $(x =_A x) \approx_{\Gamma, x:A} 1$ $(x =_A y) \approx_{\Gamma, x:A, y:A} (y =_A x)$ $p + \overline{p} \approx_\Gamma 1$ $\int_x pq \approx_\Gamma p \int_x q$ if x is not free in p

Figure 2: Basic equations for congruences on pre-polynomials. $p[x/y]$ denotes usual substitution.

A congruence is a context-indexed family of equivalence relations $\approx := (\approx_\Gamma)_\Gamma$ on pre-polynomials matching Γ, closed under associativity and commutativity of multiplication pq, neutrality of 1, every equation of Fig. 2 and such that:

■

$p \approx_\Gamma q$ implies $\overline{p} \approx_\Gamma \overline{q}$, $pr \approx_\Gamma qr$ for all r matching Γ and $p \approx_{\Gamma'} q$ for every context Γ' including the assignments of Γ, possibly in a different order;

■

$p \approx_{\Gamma, x:A} q$ implies $\int_x p \approx_\Gamma \int_x q$;

■

$p \approx_\Gamma q$ iff $p + r \approx_\Gamma q + r$ for any r matching Γ (we call this cancellativity).

Given a set S of equations of the form $p \approx_\Gamma q$ where p, q are pre-polynomials matching Γ, the congruence generated by S is the smallest congruence containing the equations of S.

In the sequel, if $\approx$ is a congruence and we write $p \approx q$, we mean $p \approx_\Gamma q$ for every Γ matching both p, q.

Lemma 8.

For any congruence $\approx$ and any pre-polynomial p, $p^2 \approx p$ iff $p\overline{p} \approx 0$. Moreover, either condition implies $\overline{p}^2 \approx \overline{p}$.

Proof.

See §A.1. ◀

$$\frac{pq \approx_{\Gamma,\Delta} 0}{p \frown_\Gamma q} \qquad \frac{}{(g^j(\vec{x}) = \vec{y}) \frown_{\vec{x}:T,\Delta} (g^{j'}(\vec{x}) = \vec{y}')}\ {\scriptstyle j \neq j'}$$

$$\frac{p \frown_{\vec{x}:T} q}{\int_{\vec{x},\vec{y}} spr \frown_\Gamma \int_{\vec{x},\vec{z}} tqr}\ {\scriptstyle (*)} \qquad \frac{p' \approx_{\Gamma,\Delta} p \quad p \frown_\Gamma q \quad q \approx_{\Gamma,\Delta} q'}{p' \frown_\Gamma q'}$$

Figure 3: Compatibility of pre-polynomials. The side condition $(*)$ is: $\Gamma, \vec{y} : U, \vec{z} : V$ (for some U, V) are domains for r, s, t, respectively, and $\vec{x}$ are the outputs of r.

Given a congruence $\approx$ over a signature $\mathfrak{S}$, we define $\approx$-compatibility to be the context-indexed family of smallest symmetric relations $\frown_\Gamma$ on monomials of $\mathfrak{S}$ closed under the rules of Fig. 3 such that $p \frown_\Gamma q$ implies that Γ is a domain for

both p and q. We write $p \frown q$ to mean $p \frown_\Gamma q$ for every Γ which is a domain for both p and q.

A $\approx$-polynomial (or simply polynomial when this is unambiguous) is a pre-polynomial p such that there exists Γ such that $q \frown_\Gamma r$ for all monomials q, r of p and such that, whenever the expression $\overline{q}$ appears in p, we have $q^2 \approx q$.

Definition 9.

An annular theory $\mathfrak{T}$ (which we abbreviate to theory in the remainder) is a triple $(\mathrm{Sort}(\mathfrak{T}), \mathrm{Gen}(\mathfrak{T}), \mathrm{Eq}(\mathfrak{T}))$ where $(\mathrm{Sort}(\mathfrak{T}), \mathrm{Gen}(\mathfrak{T}))$ is a signature and $\mathrm{Eq}(\mathfrak{T})$ is a well-founded set of equations of the form $p \approx_\Gamma q$ where p, q are pre-polynomials matching Γ, such that, calling such an equation e, we have that p and q are $\approx$-polynomials with $\approx$ the congruence generated by the equations of $\mathrm{Eq}(\mathfrak{T})$ strictly below e in the order. We denote by $\approx^{\mathfrak{T}}$ the congruence generated by $\mathrm{Eq}(\mathfrak{T})$.

Definition 10.

A $\approx$-matrix (or simply matrix) of type $\bigoplus_{k\in K} \vec{x}^k : T_k \to \bigoplus_{j\in J} \vec{y}^j : U_j$ consists of a $(J\times K)$-indexed family of $\approx$-polynomials-in-context $\left(\vec{x}^k : T_k \vdash p_{jk} :: \vec{y}^j : U_j\right)_{(j,k)\in J\times K}$ which we abbreviate to (p_{jk}), such that,

for every $j \neq j' \in J$, $k \in K$, and monomials r in p_{jk} and r' in $p_{j'k}$, $r \frown_{\vec{x}^k : T_k} r'$.(†)

We always identify $\approx$-matrices whose entries are equal modulo $\approx$.

Lemma 11.

Let (p_{jk}) and (q_{hj}) be $\approx$-matrices of type $\bigoplus_{k\in K} \vec{x}^k : T_k \to \bigoplus_{j\in J} \vec{y}^j : U_j$ and $\bigoplus_{j\in J} \vec{y}^j : U_j \to \bigoplus_{h\in H} \vec{z}^h : V_h$, respectively. Then, the family of pre-polynomials-in-context indexed by $H \times J$ defined by

$$(q \circ p)_{hk} := \left(\vec{x}^k : T_k \vdash \sum_{j\in J} \int_{\vec{y}^j} q_{hj} p_{jk} :: \vec{z}^h : V_h \right)_{(h,k)\in H\times K}$$

is a $\approx$-matrix of type $\bigoplus_{k\in K} \vec{x}^k : T_k \to \bigoplus_{h\in H} \vec{z}^h : V_h$.

Proof.

See §A.1. ◀

Let $\mathfrak{T}$ be a theory. We define a category $\mathbb{F}_1[\mathfrak{T}]$ as follows:

■

the objects are formal finite coproducts $\bigoplus_{j\in J} \vec{x}^j : T_j$ of contexts of $\mathfrak{T}$, modulo injective renaming of variables.

■

If $A := \bigoplus_{k\in K} \vec{x}^k : T_k$ and $B := \bigoplus_{j\in J} \vec{y}^j : U_j$, $\mathbb{F}_1[\mathfrak{T}](A, B)$ is the set of $\approx^{\mathfrak{T}}$-matrices of type $A \to B$.

■

Composition is given by Lemma 11, and id_A is the diagonal matrix indexed by $K \times K$ which, at position (k, k), is equal to $(\vec{x}^k =_{T_k} \vec{z}^k)$, where we used the fact that, since objects are defined up to variable renaming, by renaming each x_h^k to z_h^k, we may write A as $\bigoplus_{k\in K} \vec{z}^k : T_k$.

As defined, $\mathbb{F}_1[\mathfrak{T}]$ is not small, because the finite sets indexing the objects are arbitrary. This may be circumvented by restricting to finite subsets of a fixed, countably infinite set of indices. We leave this implicit for notational convenience, but treat $\mathbb{F}_1[\mathfrak{T}]$ as a small category.

Proposition 12.

$\mathbb{F}_1[\mathfrak{T}]$ is an ultraring.

Proof.

We spell out the required structures and properties, without entering into the details. Here "polynomial", "matrix", etc. mean $\approx^{\mathfrak{T}}$-polynomial, $\approx^{\mathfrak{T}}$-matrix, etc.

Let $A := \bigoplus_{j\in J} \vec{x}^j.T_j$ and $B := \bigoplus_{k\in K} \vec{y}^k.U_k$ be generic objects of $\mathbb{F}_1[\mathfrak{T}]$. The (strict) symmetric monoidal structure of $\mathbb{F}_1[\mathfrak{T}]$ is given by $A\otimes B := \bigoplus_{(j,k)\in J\times K} \vec{x}^j\vec{y}^k.T_j\otimes U_k$ on objects, with I as unit. On morphisms, tensor is matrix product, valid by a similar argument to the proof of Lemma 11, but without the integrals. The empty sum 0 is a zero object, the all-zero matrix of suitable size is the zero morphism between any two objects. The coproduct of A and B (where we may suppose $J\cap K = \emptyset$ and $\vec{x}^j, \vec{y}^k$ to be disjoint) is given by $\bigoplus_{l\in J\cup K} \vec{z}^l.V_l$, where $\vec{z}^l$ and V_l are equal to $\vec{x}^j$ and T_j (resp. $\vec{y}^k$ and U_k) if $l \in J$ (resp. $l \in K$). This obviously satisfies strict distributivity. The injections ι_1, ι_2 are the block column matrices $\begin{pmatrix} \mathrm{id} \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ \mathrm{id} \end{pmatrix}$, respectively, whereas the copairing of two matrices P, Q is the block row matrix $(P\ Q)$.

For what concerns the Frobenius sesquimonoid structure, δ_A is the matrix indexed over $J^2 \times J$ whose column for $j \in J$ has the element $(\vec{x}^j = \vec{y}^j)(\vec{x}^j = \vec{z}^j)$ at position (j, j) and is zero everywhere else, where we are renaming variables so that $A \otimes A = \bigoplus_{j,j'\in J} \vec{y}^j\vec{z}^{j'}.T_j \otimes T_{j'}$. ε_A is the row matrix which is equal to the monomial 1 everywhere. δ_A^* is the transpose of δ_A.

Thanks to the above definition, elementary calculations show that, if $P = (p_{kj})_{j\in J,k\in K}$ and $Q = (q_{lj})_{j\in J,l\in L}$ are matrices corresponding to morphisms $A \to B$ and $A \to C$ for some other object C (indexed by L, which we may supposed to be disjoint from K even if $C = B$), then $(P \otimes Q)\delta_A \approx^{\mathfrak{T}} 0$ is equivalent to asking that, for all $j \in J$, $k \in K$ and $l \in L$, we have $p_{kj}q_{lj} \approx^{\mathfrak{T}} 0$, which implies $p_{kj} \frown q_{lj}$, so we may form the block column matrix $R := \begin{pmatrix} P \\ Q \end{pmatrix}$ (with $K\cup L$ rows and J columns). The reader may check that the coinjections are the block row matrices (id 0) and (0 id), so R is the (unique) pairing of P, Q, as desired.

Let now $P = (p_k)_{k\in K}$ be a matrix of type $A \to I$. This means that P is a row matrix, and elementary calculations show that $(P \otimes P)\delta \approx^{\mathfrak{T}} P$ is equivalent to asking that $p_k^2 \approx^{\mathfrak{T}} p_k$ for all $k \in K$, so we are allowed to consider the polynomials $\overline{p_k}$ and define $\overline{P} := (\overline{p_k})_{k\in K}$. We need to show that $(P \otimes \overline{P})\delta \approx^{\mathfrak{T}} 0$. By similar calculations as above, this is equivalent to $p_k\overline{p_k} \approx^{\mathfrak{T}} 0$ for all $k \in K$, which holds by Lemma 8. We are left with proving that $\nabla\langle P, \overline{P}\rangle \approx^{\mathfrak{T}} \varepsilon$. Notice that, in general, $\nabla\langle Q, R\rangle = Q + R$ as matrices. This sum exists because, as shown above, $\langle Q, R\rangle$ is the block column matrix $\begin{pmatrix} Q \\ R \end{pmatrix}$. So we need to show that $p_k + \overline{p_k} \approx^{\mathfrak{T}} 1$ for all $k \in K$, but this is one of the equations of Fig. 2. ◀

3.2 Models and the Canonical Theory

The term "annular theory" in Definition 9 is not merely an analogy with logic: it is chosen so that ultraring morphisms out of $\mathbb{F}_1[\mathfrak{T}]$ become models of $\mathfrak{T}$, in a sense that we now establish.

Definition 13.

Given a signature $\mathfrak{S}$ and an ultraring $\mathcal{A}$, an $\mathfrak{S}$-structure in $\mathcal{A}$ consists of:

■

An assignment $M_0 : \mathrm{Sort}(\mathfrak{S}) \to \mathrm{ob}(\mathcal{A})$. This extends to an assignment on tensors given by $M_0(\bigotimes_i A_i) := \bigotimes_i M_0(A_i)$.

■

An assignment $M_1 : \mathrm{Gen}(\mathfrak{S}) \to \mathrm{mor}(\mathcal{A})$ such that if $g : T \to U$ then $M_1(g) : M_0(T) \to M_0(U)$.

For $\mathfrak{S}$-structures M, M' in $\mathcal{A}$, a homomorphism $m : M \to M'$ consists of a collection of morphisms $m_X : M_0(X) \to M'_0(X)$ in $\mathcal{A}$ indexed by $X \in \mathrm{Sort}(\mathfrak{S})$ such that for each $g : \bigotimes_i A_i \to \bigotimes_j B_j \in \mathrm{Gen}(\mathfrak{S})$, $M_1(g); \bigotimes_j m_{B_j} = \bigotimes_i m_{A_i}; M'_1(g)$.

Given a theory $\mathfrak{T}$ over a signature $\mathfrak{S}$ and a $\mathfrak{S}$-structure M, we can define an interpretation over M of any $\approx^{\mathfrak{T}}$-polynomial-in-context, as detailed in §A.2. The notion of model is then defined in the expected way:

Definition 14.
Let $\mathfrak{T}$ be a theory over signature $\mathfrak{S}$ and $\mathcal{A}$ an ultraring. A model of $\mathfrak{T}$ in $\mathcal{A}$ is a $\mathfrak{S}$-structure M in $\mathcal{A}$ such that for each equation $p \approx_{\Gamma,\Delta} q$ of $\mathfrak{T}$, we have $M(\Gamma \vdash p :: \Delta) = M(\Gamma \vdash q :: \Delta)$.

For models M, M' of $\mathfrak{T}$ in $\mathcal{A}$, a homomorphism $M \to M'$ is simply a homomorphism of $\mathfrak{S}$-structures. We denote by $\mathfrak{T}$-Mod($\mathcal{A}$) the category of models of $\mathfrak{T}$ in $\mathcal{A}$ and homomorphisms between them.

Proposition 15.
Let $\mathfrak{T}$ be a theory. There is an $\mathcal{A}$-natural equivalence of categories $\mathfrak{T}$-Mod($\mathcal{A}$) $\simeq$ **URing**($\mathbb{F}_1[\mathfrak{T}], \mathcal{A}$).

Proof.
See §A.2. ◀

Proposition 15 implies in particular the existence of a generic model of any annular theory $\mathfrak{T}$, corresponding to the identity functor on $\mathbb{F}_1[\mathfrak{T}]$, of which a model in any ultraring is an image by naturality.

If Proposition 12 shows us that we can generate ultrarings from theories, it may not be a surprise that every ultraring arises in this way up to equivalence.

Definition 16.
We define the canonical theory of an ultraring $\mathcal{R}$ to be the theory $\mathfrak{T}$ where:

■ $\mathrm{Sort}(\mathfrak{T}) = \mathrm{ob}(\mathcal{R})$, where we denote the sort corresponding to A by $\ulcorner A \urcorner$;

■ $\mathrm{Gen}(\mathfrak{T})$ has a generator $\ulcorner g \urcorner : \bigotimes_{j\in J} \ulcorner A_j \urcorner \to \bigoplus_{k\in K} \bigotimes_{l\in L_k} \ulcorner B_{k,l} \urcorner$ for each morphism $g : \bigotimes_{j\in J} A_j \to \bigoplus_{k\in K} \bigotimes_{l\in L_k} B_{k,l}$ in $\mathcal{R}$,[333]This is a mild abuse of notation, since the name $\ulcorner g \urcorner$ can refer to multiple generators depending on how the domain and codomain are decomposed. When not specified, the type will be the one with the minimal decomposition.

■ $\mathrm{Eq}(\mathfrak{T})$ consists of equations:

– $(\ulcorner \mathrm{id} \urcorner(\vec{x}) = \vec{y}) \approx (\vec{x} = \vec{y})$ for $\ulcorner \mathrm{id} \urcorner : \bigotimes_{j\in J} \ulcorner A_i \urcorner \to \bigotimes_{j\in J} \ulcorner A_i \urcorner$,

– $\int_{\vec{y}} (\ulcorner f \urcorner(\vec{x}) = \vec{y})(\ulcorner g \urcorner(\vec{y}) = \vec{z}) \approx (\ulcorner gf \urcorner(\vec{x}) = \vec{z})$ for composable f, g in $\mathcal{R}$ (with no coproduct decomposition of the codomains).

– $(\ulcorner \iota_j^* g \urcorner(\vec{x}) = \vec{y}) \approx (\ulcorner g \urcorner^j(\vec{x}) = \vec{y})$.

– $(\ulcorner f \otimes g \urcorner(\vec{x}, \vec{u}) = \vec{y}, \vec{v}) \approx (\ulcorner f \urcorner(\vec{x}) = \vec{y})(\ulcorner g \urcorner(\vec{u}) = \vec{v})$.

– $(\ulcorner 0 \urcorner(\vec{x}) = \vec{y}) \approx_{\vec{x}:T, \vec{y}:U, \Gamma} 0$, for any typing of $\ulcorner 0 \urcorner$.

– $(\ulcorner \delta \urcorner(\vec{x}) = \vec{x_1}, \vec{x_2}) \approx (\vec{x} = \vec{x}_1)(\vec{x} = \vec{x}_2)$

– $\ulcorner \varepsilon \urcorner(\vec{x}) \approx_{\vec{x}:T, \Gamma} 1$

–

$(\ulcorner \delta^* \urcorner(\vec{x_1}, \vec{x_2}) = \vec{x}) \approx (\vec{x}_1 = \vec{x})(\vec{x}_2 = \vec{x})$.

Theorem 17.

Let $\mathcal{R}$ be an ultraring and $\mathfrak{T}$ its canonical theory, as defined in Definition 16. Then $\mathcal{R} \simeq \mathbb{F}_1[\mathfrak{T}]$.

A.1 Proofs of Some Lemmas

Proof of Lemma 2.

We start by observing that, if $\iota_i : B_i \to B_1 \oplus B_2$ are injections, then $\iota_i \delta^*_{B_i}(\iota_i^* \otimes \mathrm{id}_{B_i}) = \delta^*_{B_1 \oplus B_2}(\mathrm{id}_{B_1 \oplus B_2} \otimes \iota_i)$. We prove this for $i = 1$, the other case is essentially identical. We omit obvious subscripts for identity arrows. Let $\chi = [\iota_1 \otimes \mathrm{id}, \iota_2 \otimes \mathrm{id}] : (B_1 \otimes B_1) \oplus (B_2 \otimes B_1) \to (B_1 \oplus B_2) \otimes B_1$ be the distributivity isomorphism. We have, using (1), $\iota_1 \delta^*_{B_1}(\iota_1^* \otimes \mathrm{id})\chi = \iota_1 \delta^*_{B_1}[\iota_1^* \iota_1 \otimes \mathrm{id}, \iota_1^* \iota_2 \otimes \mathrm{id}] = \iota_1 \delta^*_{B_1}[\mathrm{id}, 0] = [\iota_1 \delta^*_{B_1}, 0] = [\delta^*_{B_1 \oplus B_2}(\iota_1 \otimes \iota_1), \delta^*_{B_1 \oplus B_2}(\iota_2 \otimes \iota_1)] = \delta^*_{B_1 \oplus B_2}(\mathrm{id} \otimes \iota_1)\chi$, which proves the claim because χ is invertible. We will also need that, if $\kappa_i : A \otimes B_i \to (A \otimes B_1) \oplus (A \otimes B_2)$ are injections and $\chi' : (A \otimes B_1) \oplus (A \otimes B_2) \to A \otimes (B_1 \oplus B_2)$ another instance of distributivity, then $\chi' \kappa_i = \mathrm{id} \otimes \iota_i$, which holds by definition of χ'.

Suppose now that we have $f, g : A \rightrightarrows B_1 \oplus B_2$ with $\iota_i^* f = \iota_i^* g$ for $i = 1, 2$. Using the above observations, we have $\delta^*_{B_1 \oplus B_2}(f \otimes \mathrm{id})\chi' \kappa_i = \delta^*_{B_1 \oplus B_2}(f \otimes \iota_i) = \iota_i \delta^*_{B_i}(\iota_i^* f \otimes \mathrm{id}) = \iota_i \delta^*_{B_i}(\iota_i^* g \otimes \mathrm{id})$, from which, applying the first two equalities in reverse order, we obtain $\delta^*_{B_1 \oplus B_2}(f \otimes \mathrm{id})\chi' \kappa_i = \delta^*_{B_1 \oplus B_2}(g \otimes \mathrm{id})\chi' \kappa_i$ for $i = 1, 2$. Since injections are jointly epic and χ' is invertible, we infer $\delta^*_{B_1 \oplus B_2}(f \otimes \mathrm{id}) = \delta^*_{B_1 \oplus B_2}(g \otimes \mathrm{id})$, from which we conclude $f = g$ by definition δ^*-category. ◀

Proof of Lemma 8.

We start by observing that cancellativity implies $\overline{\overline{p}} \approx p$, as well as $\overline{1} \approx 0$. Next, we claim that, whenever q matches the same contexts as p and $q^2 \approx q$, we have $\overline{pq} \approx \overline{p}q + p\overline{q} + \overline{p}\,\overline{q}$. Indeed, $pq + \overline{pq} \approx 1 \approx (p + \overline{p})(q + \overline{q}) \approx pq + \overline{p}q + p\overline{q} + \overline{p}\,\overline{q}$, so we conclude by cancellativity.

Suppose that $p\overline{p} \approx 0$. Then, $p \approx p(p + \overline{p}) \approx p^2 + p\overline{p} \approx p^2$. For the converse, if $p^2 \approx p$, then using the above claim we have $\overline{p\overline{p}} = p^2 + \overline{p}^2 + \overline{p}p = p + \overline{p}(\overline{p} + p) = p + \overline{p} = 1$, so $p\overline{p} = 0$ by the first two observations. Finally, we have $\overline{p} \approx \overline{p^2} \approx p\overline{p} + \overline{p}p + \overline{p}^2 \approx \overline{p}^2$. ◀

Proof of Lemma 11.

For indices h, j, j', k, let r, r', s, s' be monomials in $p_j k, p_{j'} k, q_{hj}, q_{hj'}$, respectively. We need to check that each $(q \circ p)_{hk}$ is a polynomial by verifying that, when $j \neq j'$ we have $\int_{\vec{y}^j} sr \frown_{\vec{x}^k : T_k} \int_{\vec{y}^{j'}} s'r'$. By hypothesis, $r \frown_{\vec{x}^k : T_k} r'$, so we conclude by the last rule of Fig. 3. We must also check that the context is valid. By hypothesis, $q_{hj} p_{jk}$ is matched by $\vec{x}^k : T_k, \vec{y}^j : U_j, \vec{z}^h : V_h$, with $\vec{y}^j$ and $\vec{z}^h$ being outputs, hence each monomial $\int_{\vec{y}^j} rs$ is well-typed and matched by $\vec{x}^k : T_k, \vec{z}^h : V_h$, as required. Finally, we must check condition (†) of Definition 10. Given $h \neq h'$, we require $\int_{\vec{y}^j} sr \frown_{\vec{x}^k : T_k} \int_{\vec{y}^{j'}} t'r'$, where t' is a monomial of $q_{h'j'}$. If $j \neq j'$, we conclude as above. If $j = j'$, by hypothesis we have $s \frown_{\vec{y}^j : U_j} t'$, so we conclude by applying the last rule of Fig. 3 to these instead. ◀

A.2 Interpreting Polynomials in a $\mathfrak{S}$-structure

Let M be a $\mathfrak{S}$-structure. We start by inductively defining an interpretation over M for positive monomials-in-context, following Fig. 1.

■

As a base case, we interpret monomials over $\mathfrak{S}$ with the maximal codomain context of all outputs and the minimal domain of all remaining free variables, for each of the basic constructions of monomials:

–

$M\,(x : A \vdash (x =_A y) :: y : A)$ is $\mathrm{id}_{M_0(A)}$,

–

$M\left(\vec{x} : T \vdash (g^j(\vec{x}) = \vec{y}) :: \vec{y} : U\right)$ is $M_1(g); \iota_j^*$,

–

$M\left(\vdash 1 ::\right)$ is $\varepsilon_I = \mathrm{id}_I$,

■

Next, we describe the effect of expanding and reorganizing contexts.

–

Given $f := M(\Gamma \vdash p :: \Delta, x : A)$, let $M(\Gamma, x : A \vdash p :: \Delta) := \varepsilon_{M_0(A)} \delta^*_{M_0(A)}(f \otimes \mathrm{id}_A)$.

–

Relatedly, given $f := M(\Gamma \vdash p :: \Delta)$, we define $M(x : A, \Gamma \vdash p :: \Delta)$ to be $\beta(\varepsilon_{M_0(A)} \otimes f)$, (recalling that $\beta : I \otimes I \to I$ is the canonical isomorphism).

–

Given $M(\Gamma \vdash p :: \Delta)$, permutations of Γ and Δ are interpreted by pre- or post-composing with the corresponding symmetry morphisms.

■

Finally, we specify how monomials combine.

–

Given $f := M(\Gamma \vdash p :: \Delta, \vec{x} : T)$ and $g := M(\vec{x} : T, \Gamma' \vdash q :: \Delta')$ we can define $M\left(\Gamma, \Gamma' \vdash \int_{\vec{x}:T} pq :: \Delta, \Delta'\right)$ to be $(\mathrm{id} \otimes g)(f \otimes \mathrm{id})$; to recover an interpretation of $\int_{\vec{x}} p$, we identify this with $\int_{\vec{x}} p1$.

–

Given $f := M(x_1 : A, x_2 : A, \Gamma \vdash p :: \Delta)$ and a variable x not appearing in p, we define $M(x : A, \Gamma \vdash p[x_1/x][x_2/x] :: \Delta)$ to be $f(\delta_{M_0(A)} \otimes \mathrm{id})$.

–

Dually, given $f := M(\Gamma \vdash p :: x_1 : B, x_2 : B, \Delta)$ and a variable x not appearing in p, we define $M(\Gamma \vdash p[y_1/y][y_2/y] :: y : B, \Delta)$ to be $(\delta^*_{M_0(B)} \otimes \mathrm{id})f$.

Let us check that the above interpretations are well-defined. The first set of rules give unambiguous interpretations. For the second set, there is exactly one way to arrive a given context using these rules up to reordering, so it suffices to check that the order of operations does not affect the result; this is straightforward, relying on the fact that δ^* is symmetric.

The only place where ambiguity is introduced in the third set of rules, where the interpretation ostensibly depends on how the domain and codomain contexts are partitioned. We take these case by case.

■

Consider $M\left(\Gamma, \Gamma' \vdash \int_{\vec{x}:T} pq :: \Delta, \Delta'\right)$. By disjointness, Δ and Δ' must respectively contain only outputs of p and q, so this is unambiguous; the remaining free variables of p and q must respectively go into Γ and Γ'. Any remaining free variables, whether assigned to Γ or Γ', must be interpreted by ε, which eliminates any remaining ambiguity.

■

In $M(x : A, \Gamma \vdash p[x_1/x][x_2/x] :: \Delta)$, there is formally a choice to make about how to assign the instances of x to x_1 or x_2; however, since the inductive construction will eventually disambiguate between any pair of repeated instances, commutativity and associativity of δ guarantee that any such choice will result in the same interpretation (considering also that the interpretation is invariant under α-equivalence).

■

Similarly, distinguishing duplicates of variables in the codomain produces a well-defined result by commutativity and associativity of δ^*, so we are done.

Next, given $f := M(\Gamma \vdash p ::)$, we have that $M\left(\Gamma \vdash p^2 ::\right)$ is exactly $(f \otimes f)\delta$. As such, when these interpretations are equal, we may take $M(\Gamma \vdash \overline{p} ::)$ to be $\overline{f}$ provided by Definition 3, thus extending the above to monomials featuring $\overline{p}$.

Finally, we define $M(\Gamma \vdash 0 :: \Delta)$ to be the zero morphism of the appropriate type.

Lemma 23.

Suppose we are given a theory $\mathfrak{T}$ over a signature $\mathfrak{S}$ and a $\mathfrak{S}$-structure M in $\mathcal{A}$. Suppose that whenever we have positive monomials-in-context $\Gamma \vdash p :: \Delta$ and $\Gamma \vdash q :: \Delta$ such that $p \approx_{\Gamma,\Delta} q$, it is the case that $M(\Gamma \vdash p :: \Delta) = M(\Gamma \vdash q :: \Delta)$, so that in particular M can interpret all monomials which are $\approx$-polynomials. Suppose further that M identifies p and q whenever $p \approx q$ when p and q are either monomials or 0. Then $p \frown_{\mathfrak{T}} q$ implies the pairing $\langle p, q\rangle$ exists in $\mathcal{A}$.

Proof.

It suffices to check the axioms of Fig. 3 which determine the compatibility relation. Indeed, $pq \approx 0$ guarantees that if M respectively interprets p, q as f, g in a given context then the first axiom of ultrarings Definition 3 ensures that the pairing exists.

The pairing of two components j, j' of (the interpretation of) a generator g is the result of coinjecting g onto the coproduct of their codomains.

If $p \frown q$ and $\langle p, q\rangle$ exists, we can obtain the pairing of the interpretations of $\int_{\vec{x},\vec{y}} spr$ and $\int_{\vec{x},\vec{z}} tqr$ by precomposing with r and postcomposing with $s \oplus t$.

The existence of pairings in the last instance is immediate, so we are done. ◀

As such, given $\mathfrak{T}$ and a model M satisfying the conditions of Lemma 23, M has a well-defined interpretation of any $\approx$-polynomial-in-context $\Gamma \vdash p :: \Delta$ as the joint pairing of the monomials appearing in p (with the same context), followed by the codiagonal map.

Proposition 15. [Restated, see original statement.]

Let $\mathfrak{T}$ be a theory. There is an $\mathcal{A}$-natural equivalence of categories $\mathfrak{T}\text{-Mod}(\mathcal{A}) \simeq \mathbf{URing}(\mathbb{F}_1[\mathfrak{T}], \mathcal{A})$.

Proof.

We have thus far shown that a $\mathfrak{T}$-model M in $\mathcal{A}$ uniquely determines interpretations of $\approx^{\mathfrak{T}}$-polynomials-in-context. To produce a functor $\mathbb{F}_1[\mathfrak{T}] \to \mathcal{A}$, we need to extend this to $\approx^{\mathfrak{T}}$-matrices. Each entry of a matrix (p_{jk}) determines a morphism $M_0(T_k) \to M_0(U_j)$. Condition (†) of Definition 10 ensures that for fixed k, the interpretations of p_{jk} can be paired to give a morphism $M_0(T_k) \to \bigoplus_{j\in J} M_0(U_j)$; via the universal property of the coproduct, these produce a morphism $\bigoplus_{k\in K} M_0(T_k) \to \bigoplus_{j\in J} M_0(U_j)$. This extended construction ensures that the monoidal product and coproduct, as well as δ, δ^* and ϵ, are preserved.

Conversely, a morphism of ultrarings $F : \mathbb{F}_1[\mathfrak{T}] \to \mathcal{A}$ defines a model M by its restriction to the sorts and generators. The equations of $\mathfrak{T}$ necessarily hold because they correspond to morphisms forced to be equal in $\mathbb{F}_1[\mathfrak{T}]$, so are mapped by F to the equal morphisms in $\mathcal{A}$.

It is straightforward to verify that a model morphism uniquely determines a monoidal natural transformation, and conversely, since the latter are determined by their components at objects which generate the category under monoidal products and coproducts.

Finally, naturality in $\mathcal{A}$ follows from the fact that morphisms of ultrafunctors preserve the required structure: composing with an ultraring morphism $\mathcal{A} \to \mathcal{B}$ turns a $\mathfrak{T}$-model in $\mathcal{A}$ into one in $\mathcal{B}$. ◀

Theorem 17. [Restated, see original statement.]

Let $\mathcal{R}$ be an ultraring and $\mathfrak{T}$ its canonical theory, as defined in Definition 16. Then $\mathcal{R} \simeq \mathbb{F}_1[\mathfrak{T}]$.

Proof.

We have a functor $\ulcorner - \urcorner : \mathcal{R} \to \mathbb{F}_1[\mathfrak{T}]$ sending each object X to the sort $\ulcorner X \urcorner$ and each morphism $g : X \to Y$ to the corresponding generator (with no decomposition of X and Y as tensors or coproducts). We show that $\ulcorner - \urcorner$ is an equivalence of ultrarings.

Consider the generators (which we distinguish by a subscript) $\ulcorner \mathrm{id} \urcorner_1 : \ulcorner \bigotimes_{j\in J} A_j \urcorner \to \bigotimes_{j\in J} \ulcorner A_j \urcorner$ and $\ulcorner \mathrm{id} \urcorner_2 : \bigotimes_{j\in J} \ulcorner A_j \urcorner \to \ulcorner \bigotimes_{j\in J} A_j \urcorner$. By the second and first equations, these are mutual inverses in $\mathbb{F}_1[\mathfrak{T}]$. Moreover, naturality of these morphisms follows from the fourth axiom (applied inductively), so $\ulcorner - \urcorner$ preserves the monoidal product.

Consider the generator $\ulcorner \mathrm{id} \urcorner : \ulcorner \bigoplus_{j\in J} A_j \urcorner \to \bigoplus_{j\in J} \ulcorner A_j \urcorner$. By the third equation, we have that $\left(\ulcorner \mathrm{id} \urcorner^j(\vec{x}) = \vec{y}\right) \approx (\ulcorner \iota_j^* \urcorner(\vec{x}) = \vec{y})$, so this is equal to the row vector $(\ulcorner \iota_{j_1}^* \urcorner, \dots, \ulcorner \iota_{j_n}^* \urcorner)$.

Dually, we have generators $\ulcorner \iota_i \urcorner : \ulcorner A_i \urcorner \to \bigoplus_{j\in J} \ulcorner A_j \urcorner$. By the third, first and fifth equations, we have $\left(\ulcorner \iota_i \urcorner^j(\vec{x}) = \vec{y}\right) \approx \left(\ulcorner \iota_j^* \iota_i \urcorner(\vec{x}) = \vec{y}\right)$ which gives $(\vec{x} = \vec{y})$ if $i = j$ and 0 otherwise. Thus $\ulcorner - \urcorner$ preserves the coproduct injections; in particular, these morphisms are compatible, so we can construct the column vector $(\ulcorner \iota_{j_1} \urcorner, \dots, \ulcorner \iota_{j_n} \urcorner)^t$. We can use the second and third axioms to conclude that this column vector is the inverse to $\ulcorner \mathrm{id} \urcorner$, so $\ulcorner - \urcorner$ preserves coproducts. Formally, we must also check that the universal morphisms are respected by this isomorphism, in the sense that $\int_{\vec{y}} (\ulcorner [f_1, \dots, f_n] \urcorner(\vec{x}) = \vec{y}) (\ulcorner \mathrm{id} \urcorner(\vec{y}) = \vec{z}) \approx \int_{\vec{y}'} (\ulcorner \mathrm{id} \urcorner(\vec{x}) = \vec{y}') ((\ulcorner f_1 \urcorner, \dots, \ulcorner f_n \urcorner)(\vec{y}') = \vec{z})$. It suffices to check this componentwise by composing with ι_i, which works by the preceding observations and the second axiom.

For the δ^*-category structure, we observe that the sixth, seventh and eighth axioms directly ensure that these are preserved up to the isomorphisms constructed above. With this, we have shown that $\ulcorner - \urcorner$ is a valid morphism of ultrarings.

Conversely, there is a canonical model of the canonical theory $\mathfrak{T}$ in $\mathcal{R}$, given by interpreting $\ulcorner A \urcorner$ as A and $\ulcorner g \urcorner$ as g (with the relevant typing). The claim that this $\mathfrak{S}$-structure is indeed a model amounts to saying that the interpretations of the monomials appearing in the axioms of Definition 16 are equal in $\mathcal{R}$, which is straightforward to check.

Finally, to see that $\ulcorner - \urcorner$ is an equivalence, we simply observe on the one hand that the induced model in $\mathbb{F}_1[\mathfrak{T}]$ is isomorphic to that corresponding to the identity functor and on the other that the induced endofunctor of $\mathcal{R}$ is naturally isomorphic to the identity. The involved natural isomorphisms are the structural ones constructed above. ◀

References

References[1] Samson Abramsky, Anuj Dawar, and Pengming Wang. The pebbling comonad in finite model theory. In Proceedings of LICS, pages 1–12. IEEE Computer Society, 2017. doi:10.1109/LICS.2017.8005129. [2] Samson Abramsky and Luca Reggio. Arboreal categories: An axiomatic theory of resources. Log. Methods Comput. Sci., 19(3), 2023. doi:10.46298/LMCS-19(3:14)2023. [3] Samson Abramsky and Nihil Shah. Relating structure and power: Comonadic semantics for computational resources. J. Log. Comput., 31(6):1390–1428, 2021. doi:10.1093/LOGCOM/EXAB048. [4] John C. Baez, Joe Moeller, and Todd Trimble. 2-rig extensions and the splitting principle. arXiv:2410.05598 [math.CT], 2024. [5] Lenore Blum, Mike Shub, and Steve Smale. On a theory of computation and complexity over the real numbers. Bulletin of the American Mathematical Society, 21(1):1–46, 1989. [6] Martin Brandenburg. Tensor categorical foundations of algebraic geometry. Ph.d. thesis, University of Münster, 2014. [7] Martin Brandenburg and Alexandru Chirvasitu. Tensor functors

between categories of quasi-coherent sheaves. Journal of Algebra, 399:675–692, 2014. [8] Aurelio Carboni, Stephen Lack, and R.F.C. Walters. Introduction to extensive and distributive categories. Journal of Pure and Applied Algebra, 84(2):145–198, 1993. [9] Baptiste Chanus and Damiano Mazza. A categorical approach to describing complexity classes. In preparation. Available on the authors' web page, 2024. [10] Alexandru Chirvasitu and Theo Johnson-Freyd. The fundamental pro-groupoid of an affine 2-scheme. Appl. Categ. Structures, 21(5):469–522, 2013. doi:10.1007/S10485-011-9275-Y. [11] Anuj Dawar, Tomás Jakl, and Luca Reggio. Lovász-type theorems and game comonads. In Proceedings of LICS, pages 1–13. IEEE, 2021. doi:10.1109/LICS52264.2021.9470609. [12] Nikolai Durov. New Approach to Arakelov Geometry. arXiv 0704.2030 [math.AG], 2007. [13] Nikolai Durov. Classifying vectoids and operad kinds. Proc. Steklov Inst. Math., 273:48–63, 2011. [14] Ronald Fagin. Generalized first-order spectra and polynomial-time recognizable sets. In Complexity of Computation, SIAM-AMS Proccedings, volume 7, pages 43–73, 1974. [15] Erich Grädel and Klaus Meer. Descriptive complexity theory over the real numbers. In Proceedings of STOC, pages 315–324. ACM Press, 1995. doi:10.1145/225058.225151. [16] Neil Immerman. Relational queries computable in polynomial time. Information and Control, 68(1):86–104, 1986. doi:10.1016/S0019-9958(86)80029-8. [17] Neil Immerman. Descriptive complexity. Graduate texts in computer science. Springer, 1999. doi:10.1007/978-1-4612-0539-5. [18] Bart P. F. Jacobs. Categorical Logic and Type Theory, volume 141 of Studies in logic and the foundations of mathematics. North-Holland, 2001. [19] Peter T. Johnstone. Sketches of en Elephant. A Topos Theory Compendium. Volume 2. Oxford University Press, 2002. [20] Melven R. Krom. The decision problem for a class of first-order formulas in which all disjunctions are binary. Zeitschrift für Mathematische Logik und Grundlagen der Mathematik, 13(1–2):15–20, 1967. [21] Saunders Mac Lane and Ieke Moerdijk. Sheaves in Geometry and Logic. Springer, 1994. [22] Yoàv Montacute and Glynn Winskel. Concurrent games over relational structures: The origin of game comonads. In Proceedings of LICS, pages 58:1–58:14. ACM, 2024. doi:10.1145/3661814.3662075.